

Analysis of the Systematic Destabilization of Tigray Through Violent Displacement and Agricultural Destruction

Project Focus and Purpose

This project assesses the ongoing humanitarian crisis in Tigray through a geographical and contextual analysis of internally displaced people (IDPs), food insecurity, medical needs, health facility functionality, market functionality, and regional access constraints. Using sub-regional mapping in conjunction with post-conflict reports, the analysis aims to convey how these indicators interact spatially and how they reflect broader patterns of civilian harm, livelihood collapse, and systemic destabilization within Tigray.

The focus of the analysis was not only to visualize humanitarian needs but also to understand how displacement, collapse of basic services, and agricultural destruction operate together as mutually reinforcing mechanisms. The maps were designed to show where stress is concentrated, while the written analysis explains why these patterns exist and how they affect the civilian population over time.

Spatial analysis was conducted in QGIS using zone-level administrative boundaries. Prior to mapping, datasets were analyzed in Python using pandas to rank regions according to severity across food and medical needs and access constraints, structure region-level indicators, check for missing values, and create derived metrics (such as displacement rate per population). Python was also used to create summary tables and bar-chart visualizations to confirm patterns before spatial joins in QGIS. Cleaned and organized metrics were then merged in QGIS using standardized region/zone identifiers. Next, relative intensity rather than absolute accuracy was visualized using graduated symbology.

II. Internally Displaced Persons (IDPs): Scale, Distribution, and Impact

Data from humanitarian reports indicates that displacement in Tigray reached unprecedented levels following the outbreak of conflict. “By May 2021, the number of internally displaced persons in Tigray had reached approximately 1.9 million, and by the end of 2021, displacement had doubled to an estimated 4.2 million people due to the ongoing conflict.” (GEOGLAM Crop Monitor – Conflict and Food Insecurity Report (2022)). These numbers reflect the severity and geographic and systematic entrapment of the crisis, as they show displacement that mostly occurs within Tigray itself rather than cross-border movement.

With greater concentrations in the western and central zones, the IDP concentration map illustrates the uneven displacement distribution throughout Tigray. This can be seen in QGIS by zone-level graduation, which aligns spatially with areas of restricted access and service disruption and shows higher IDP concentration in western and central zones. Data from these

geographic patterns indicates that displacement is a multi-dimensional force: it represents the endpoint of a humanitarian emergency while simultaneously acting as the starting point for further systemic instability. “Large-scale displacement has significantly disrupted livelihoods and agricultural activities, as households fled rural areas and abandoned productive assets, contributing to increased pressure on host communities and local systems.” (GEOGLAM Crop Monitor – Conflict Reports), which creates immense strain on already weakened health systems.

In Tigray, displacement must be viewed as both a contributing factor and an outcome of a larger systemic breakdown. Households abandon their farmland, livestock, tools, and stored harvests when they are driven from their homes and land by invading forces or fleeing violence. Long-term displacement weakens adaptation skills, increases reliance on aid, and hinders communities' ability to recover even in the event that the conflict stops. “Displacement has resulted in widespread abandonment of farmland, loss of livestock and tools, and a reduction in household coping capacity, limiting the ability of affected populations to recover even when active fighting subsides.” (GEOGLAM Crop Monitor – Conflict Reports).

III. Agricultural Sector Destabilization in Tigray

Agriculture has historically been the backbone of livelihoods in Tigray. “The majority of households in Tigray depend on rainfed subsistence agriculture and livestock production as their primary source of food and income.” (GEOGLAM Crop Monitor – Ethiopia Context Sections). Therefore, the effects of conflict-related disruption to this sector extend well beyond short-term food shortages and have an impact on the region's long-term social stability and economic viability. According to reports, Tigray's agricultural production was systematically undermined by a combination of forced displacement, limitations on travel, and direct attacks on farming systems, in addition to being disrupted by fighting.

Due to insecurity and labor shortages, crops were reportedly looted, burned, or allowed to rot in fields while farmers were barred from plowing or harvesting. According to a report from a Crop Monitor Conflict Report, they stated, “Additionally, in a June 2021 news report, farmers, aid workers, and local officials noted that farmers had been stopped from ploughing or harvesting, seeds for planting had been stolen, farm equipment had been looted, and livestock had been killed. Another report noted direct instances of crop pillaging and burning. Women and children were also reported to have worked the lands during the daytime, while adult men had to remain in the rural villages working at night to avoid being a target.” (GEOGLAM Crop Monitor – Conflict Report, June 2021 references).

The destruction extended to agricultural resources and infrastructure. Seed stores, tools, irrigation equipment, and fertilizer supplies were reportedly looted or destroyed, while agricultural service points and warehouses were vandalized or shut down. This resulted in the collapse of local seed systems and supply chains, leaving farmers unable to resume production even in areas where fighting temporarily subsided. “Agricultural services and available supply

systems were severely disrupted as warehouses and service centers were vandalized or closed, resulting in limited availability of seeds, fertilizer, and other essential inputs.” (GEOGLAM Crop Monitor – Conflict Reports)

Looking at the agricultural collapse it was further escalated by losses of livestock. Cattle, oxen, and small ruminants, which are necessary not only for food but also for plowing and revenue generation, are frequently slaughtered or stolen, according to reports. In particular, the loss of oxen severely restricted the households' ability to cultivate land, preventing many farmers from participating in upcoming cultivating seasons. “Widespread loss of livestock, including plough oxen, significantly constrained households’ ability to cultivate land and reduced agricultural production across multiple seasons.” (GEOGLAM Crop Monitor – Conflict Reports)

These patterns indicate that agricultural harm in Tigray was not limited to incidental damage from conflict but followed repeated, overlapping pathways that stripped communities of productive capacity. When fields are abandoned, inputs destroyed, livestock removed, and labor displaced simultaneously, the result is a sustained inability to produce food rather than a temporary shock. “The repeated destruction of crops, livestock, agricultural inputs, and services has had a cumulative impact on production capacity, indicating sustained damage rather than temporary disruption.” (GEOGLAM Crop Monitor – Conflict Reports).

IV. Market Functionality and Economic Strain

Although some markets in Tigray continued to operate, the functional markets layer shows that market functionality differed greatly by zone. Crucially, market functionality refers to the actual existence of exchange activity under limited circumstances rather than affordability or food security.

The effects of the agricultural collapse reduced supply even in zones where markets continued to function; households faced severe limitations due to loss of income, high prices, and restricted movement. It is vital to be able to understand the situation on the ground. “The presence of functioning markets does not necessarily translate into food access for households, as loss of income, high prices, and movement restrictions continue to limit purchasing ability.” (GEOGLAM Crop Monitor – Conflict Reports).

Markets also became sites of stress as “Concentrations of displaced people increased demand in certain locations without corresponding increases in supply, intensifying competition for food and basic goods.” (GEOGLAM Crop Monitor – Conflict Reports), further marginalizing displaced households and host communities alike. These economic constraints further compounded medical vulnerability, as households faced reduced access to both food and health services simultaneously.

V. Health System Disruption and Medical Need

Prior to the conflict Tigray excelled in having one of the strongest healthcare systems in Ethiopia, with an extensive infrastructure that consists of over 750 health posts, 220 health centers, 40 hospitals, and advanced facilities, and a fleet of around 250 ambulances.

While having such a strong system in place, Tigray also had a well built referral system with ambulances, and significant growth in the private sector. Which contributed to good healthcare results with maternal child health, vaccinations, and a large workforce within the healthcare system. Thus being able to provide basic health care accessibility to most of the residents before the devastating war which resulted in its near-total collapse.

This collapse of the regional health system is intertwined with wide-range patterns of violence and obstruction. As displacement worsened and agricultural livelihoods were devastated, the need for medical assistance rose sharply amid a health infrastructure which had already been severely weakened.

The systemic destruction that occurred in Tigray resulted in an extensive humanitarian catastrophe, the collapse of the healthcare system, and the mass systemic destruction of healthcare resources. Before the war, over 90 percent of public and private health facilities in Tigray were physically accessible, undermining the scale of institutional loss following the beginning of hostilities (Gufue et al., 2024).

According to reports from Doctors Without Borders, “the two-year Tigray conflict, characterized by extreme violence, widespread looting, and massive displacement, resulted in the near-total collapse of the health care system. Health facilities lacked medical supplies, biomedical equipment, and health workers, and what health workers were available were often unpaid” (Doctors Without Borders / MSF, 2023).

Since the start of the conflict in November 2020, millions of people have been internally displaced, worsened widespread food insecurity, the intentional destruction of medical facilities, and the displacement of Tigrayan refugees into neighboring Sudan (Gufue et al., 2024).

This resulted in a major spike in severe malnutrition, untreated chronic illness, maternal health issues, and trauma-related injuries which placed overwhelming stress on a system that is operating far below its capacity, this highlights how the destruction of the infrastructure resulted in widespread suffering and death.

The breakdown extended beyond the facilities to targeted threats against humanitarian personnel. An internal review done by Médecins Sans Frontières found that the June 24, 2021 killing of three clearly identified MSF staff in central Tigray was “an intentional and targeted killing of clearly identified aid workers” and established that “a convoy of Ethiopian National

Defense Forces (ENDF) was present at the time of the incident, on the same road where the MSF personnel were killed” (Médecins Sans Frontières, 2025).

The review further reported that witness accounts “directly implicated ENDF soldiers in the attack” through detailed observations, including an alleged order over military radio to shoot at the approaching MSF vehicle. (Doctors Without Borders).

Several investigations reported additional unprovoked widespread violence inflicted against humanitarian workers operating in the region, including 23 aid workers (including three United Nations Security Guards) killed during the course of the war. In addition, these events also caused a deferment or suspension of operational services from humanitarian agencies, which decreased civilian access to life-saving treatment. These figures include the figures from the MSF attack.

Looking at the health infrastructure in Tigray it clearly suffered significantly, according to assessments conducted by UNICEF and the Tigray Health Bureau. Only a small percentage of the 263 facilities evaluated were fully operational, with 205 health centers (78%) and 24 hospitals (60%) suffering partial damage.

Due to a shortage of medical supplies, equipment, and qualified personnel, only 22% of the 186 facilities deemed functional were fully equipped to provide care, 19% were only partially operational, and 59% were completely non-operational (Gufue et al., 2024).

Out of the 230 health centers only 29 were still open at the height of the crisis, and only 1% of them could provide full services concerning gender-based violence, depriving thousands of vulnerable people of vital treatment (Gufue et al., 2024). These figures reflect tens of thousands of communities cut off from lifesaving treatment, where illness, injury, and trauma went largely untreated.

The data reflects not only the loss of individual lives but also how the threat to humanitarian actors accelerated the withdrawal and suspension of critical medical operations and staff throughout the region, which resulted in deepening gaps for civilian access to care and further contributing to an intensified humanitarian crisis. In instances where health facilities remained infrastructurally sound, road insecurity, fuel shortages, and intentional travel restrictions severely limited civilian access and impeded the delivery of medical supplies.

Another MSF assessment found that of 106 facilities visited between December 2020 and March 2021, nearly 70 percent had been looted, more than 30 percent had been damaged, and only 13 percent were fully operating, displaying the aggravated effects of violence and access constraints on survival systems and humanitarian response.

Besides infrastructural damage, the conflict included dozens of accounts of atrocities, massacres, and overwhelming mass displacement, these actions are widely regarded as genocidal and ethnic cleansing which were done upon the people of Tigray. Upwards of 91 percent of the population required humanitarian assistance, with extensive documented hunger-related deaths and hundreds of thousands experiencing extreme food scarcity within months of shifting territorial control (UNOCHA 2024).

While several studies demonstrated that the conflict significantly undermined Tigray's healthcare system, very few have analyzed the correlated financial consequences, thereby emphasizing the need to evaluate direct economic damage and broader inflicted destruction of the public health sector (Gufue et al., 2024).

VI. Systematic Destabilization and Civilian Impact

When considered holistically, the geographical trends of IDP concentration, market disruption, agricultural collapse, health system failure, and access restrictions indicate an act of systematic destabilization involving extensive civilian harm, including acts that human rights and humanitarian organizations have documented as extrajudicial violence rather than isolated humanitarian shocks. These systems operate in ways that compromise the foundation of survival and exacerbate suffering among civilians.

Documented patterns of destruction of civilian infrastructure offer validity to this conclusion. “civilian structures in towns in Tigray, including hospitals, schools, factories, and businesses, were shelled, looted, and destroyed by Ethiopian federal forces and regional militias, and by Eritrean armed forces” (Human Rights Watch, 2023). These repeated patterns across sectors and regions demonstrate consistency in impact over time, a defining characteristic of systematic harm rather than sporadic conflict damage.

The destruction of agriculture removes food and income, displacement removes labor and stability, market disruption removes purchasing power, and health system collapse removes the ability to survive illness and injury. When these occur simultaneously and persist over time, they function as a comprehensive assault on civilian livelihoods.

For the population of Tigray, these dynamics translate into prolonged hunger, dependency, health deterioration, and psychological trauma. The strain extends beyond immediate survival, affecting social cohesion, generational recovery, and the long-term capacity of communities to rebuild.

VII. Limitations and Ethical Considerations

The challenge of evaluating data in an ongoing war zone, reporting gaps, and data availability all constrain this investigation. Instead of full or continuous coverage, geographical information represents approximations and time fragments.

Despite these drawbacks, the findings' reliability is bolstered by the integration of independent data sources and comparable geographical trends. In example, while depicting displacement and access limitations, careful consideration was made to avoid suggesting ideal conditions where harm is evident.

VIII. Conclusions

This analysis shows that individual data alone are inadequate to comprehend the vast humanitarian issue in Tigray. The breakdown of the health system, market disruption, agricultural destruction, and displacement all work together as reinforcing forces that have devastated civilian life.

The research highlights how humanitarian data can show where needs are most pressing as well as how those needs are created and maintained by integrating geographical analysis with contextual reporting. The results highlight how crucial it is to see Tigray's crisis as a systemic failure of livelihoods and means of survival, with long-lasting effects.